

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	37.6	16.3	3	1	0.7	41.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	39.6	13.9	3	0.2	0.1	43.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.3	17.2	3.3	0.5	0.8	41.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	39	14.7	2.8	0.2	0.2	43.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.7	16.7	3.1	0.8	0.7	42
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.4	14.6	2.6	0.1	0.2	44.1
U2 - Efficacy additive alpha prior Beta(1, 1)	36.9	17	2.9	0.9	0.9	41.4
U3 - Efficacy additive alpha prior Beta(1, 1)	38.7	14.2	2.5	0.3	0.2	44.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	6.5	2.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.4	6.2	2.7	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.9	6.6	2.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.4	6.2	2.7	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	6.6	2.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.4	6.2	2.7	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	6.5	2.9	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.4	6.2	2.7	1.7	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.4	0.8	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.2	0.8	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.3	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.2	0.8	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	0.8	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.1	0.8	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.3	0.8	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.2	0.8	0.7	0.2	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.6	17.2	38.5	14.1	3.7	0.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.8	17.4	34.9	12.3	2.6	1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.6	16.7	37.5	15.1	3.2	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.1	17	35.1	12.5	2.4	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27	17	37.3	14.3	3.5	0.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.1	18.4	33.5	12.8	2.2	1
U2 - Efficacy additive alpha prior Beta(1, 1)	27.1	16.3	37.3	14.7	3.7	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	32.3	17.5	34.6	12.4	2.3	0.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	6.8	8.8	4.1	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	6.9	8.7	3.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	6.7	8.8	4.2	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	6.9	8.7	3.9	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	6.8	8.8	4.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	6.9	8.7	4	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	6.7	8.9	4.2	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	6.8	8.7	3.9	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.6	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.1	1.6	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.6	0.9	0.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	44.7	26.8	12.1	3	2.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.7	50.2	22	9.9	1.8	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	45.6	26.3	12.6	2.5	2.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.6	51.4	22	9.2	1.5	2.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	45.6	26.7	12	2.9	2.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.5	49.3	24	9.3	1.6	2.3
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	45.4	26.2	13	2.7	2.2
U3 - Efficacy additive alpha prior Beta(1, 1)	13.4	51.7	20.9	10.4	1.4	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.3	10.6	7.2	4.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.5	11.5	6.7	3.8	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.3	10.5	7.2	4.3	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.5	11.5	6.7	3.9	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.3	10.6	7.1	4.3	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.5	11.5	6.8	3.8	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.3	10.5	7.1	4.3	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.5	11.4	6.7	3.9	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.8	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.8	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.8	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.6	1.1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	6.1	15.9	50.9	19.7	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	7.5	14.4	49.9	16.2	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.6	5.2	16.5	51.1	19.4	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	7.6	15.4	49.3	15.7	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.3	5.2	16.9	50.8	19.6	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	7.2	14.2	50.9	16	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	7.6	5.1	16.6	50.7	19.8	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	7.5	14.5	49.7	16.6	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	4.3	6.4	10.3	4.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.4	6.5	10.5	4.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	4.3	6.5	10.4	4.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.4	6.6	10.5	4.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	4.3	6.5	10.3	4.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	4.5	6.5	10.5	4.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	4.3	6.5	10.3	4.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.3	4.4	6.6	10.5	4.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.5	2.4	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.5	2.4	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.5	2.4	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.5	2.5	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.5	2.3	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	32.8	27.7	21.4	10.9	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	39.3	25.3	18.1	6.9	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	31.3	29.7	21.5	10.7	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	40.9	24.5	17.2	7.4	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.5	32	28.5	21.6	11	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	39.2	25.5	18.2	6.9	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	6.7	32.6	28	21	11.3	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	10	40.8	23.7	18.2	6.9	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	6.6	7.1	7.4	5.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	7.5	7.1	7.1	4.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	6.6	7.1	7.3	5.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	7.5	7.2	7	4.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	6.7	7.1	7.2	5.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	7.5	7.2	7	4.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.7	6.7	7.1	7.4	5.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	7.4	7.2	7	4.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.9	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1.9	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1.9	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1.9	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1.9	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	61.7	18.3	5.5	2	0.5	12
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.3	16.1	3.6	2	0.2	11.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	62.4	17.3	5.9	2.3	0.5	11.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.8	15.6	3.5	1.7	0.2	12.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	61.4	17.8	5.7	2.2	0.6	12.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	66.3	15.8	3.8	1.7	0.2	12.2
U2 - Efficacy additive alpha prior Beta(1, 1)	62.6	16.9	6	2.1	0.5	11.9
U3 - Efficacy additive alpha prior Beta(1, 1)	65.6	17	3.4	2	0.2	11.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	6.2	3	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	6	2.6	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.8	6.2	3	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	6	2.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.7	6.2	3	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	6	2.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.8	6.1	3	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.4	6.1	2.7	1.4	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.6	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.4	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.5	0.8	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.8	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.8	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.7	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	44.7	24.3	12.4	4.5	0.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	48.7	21.8	9.6	2.4	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	44.8	23.4	12.9	4.7	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	49.1	21.6	10.1	2.4	1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	44.8	23.2	13.3	4.7	0.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.9	48.1	21.8	9.7	2.4	1.1
U2 - Efficacy additive alpha prior Beta(1, 1)	13.1	43.9	24	13.4	4.6	1
U3 - Efficacy additive alpha prior Beta(1, 1)	16.8	48.1	21.8	10.2	2	1.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	11.7	6.1	3.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.7	12.5	5.9	2.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.4	11.7	6.2	3.3	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.6	12.5	5.9	2.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	11.6	6.2	3.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.7	12.5	5.8	3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	11.6	6.1	3.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.7	12.5	5.8	2.9	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.3	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.4	1.2	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.3	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.4	1.2	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.3	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.4	1.2	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.3	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.3	1.2	0.6	0.2	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.7	15.1	2.8	1.5	0.7	33.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	47.2	14	3.1	1	0.8	33.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.4	14.9	3.7	1.4	0.7	32.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	47.6	13.7	3.2	1.2	0.7	33.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.7	15.3	3.3	1.5	0.7	33.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	47.2	14.5	2.9	1.2	0.7	33.5
U2 - Efficacy additive alpha prior Beta(1, 1)	46.2	15.1	3.4	1.5	0.7	33.1
U3 - Efficacy additive alpha prior Beta(1, 1)	46.4	14.6	2.9	1.3	0.7	34.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15	6.2	2.7	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	6	2.5	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	6.2	2.7	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.5	6	2.5	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.2	6.2	2.6	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	6	2.5	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	15	6.2	2.7	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	6	2.5	1.4	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.9	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	2.8	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.9	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.7	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.9	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.8	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.9	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	2.8	0.7	0.5	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	13.1	21.4	14.7	32.4	3.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.5	15.1	18.9	13.1	26.5	3.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	13.2	20.4	15.4	32.2	3.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.5	16	18.6	13.8	26.1	4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	12.1	21.9	15.6	31.2	3.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.3	15.3	20.2	12.6	26.7	3.9
U2 - Efficacy additive alpha prior Beta(1, 1)	15.8	13.4	20.5	15.6	31	3.7
U3 - Efficacy additive alpha prior Beta(1, 1)	22.2	15.9	18.7	13.8	25.6	3.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.6	5.7	5.9	5.4	6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	5.6	5.8	5.4	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.6	5.8	5.9	5.4	5.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	5.6	5.8	5.4	5.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	5.7	5.9	5.4	6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.2	5.6	5.8	5.4	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.6	5.7	5.8	5.5	6	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.2	5.6	5.8	5.5	5.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.6	1.9	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.3	1.6	1.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.6	2	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.3	1.6	1.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.5	1.9	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	1.6	1.9	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.4	1.5	2	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.3	1.6	1.9	0.9	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	41.9	17.9	7.6	3	18.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	44.7	16.1	6.4	1.5	18.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	42.6	17.7	7.2	3	18.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	45.3	16	6.4	1.9	18.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	44.3	16.9	6.8	2.6	17.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	45.3	16	5.7	1.8	18.3
U2 - Efficacy additive alpha prior Beta(1, 1)	11.1	42.5	17.9	7	3	18.5
U3 - Efficacy additive alpha prior Beta(1, 1)	12.8	44.9	16.8	5.4	1.9	18.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	10.3	5.3	2.7	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	11	5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	10.4	5.3	2.6	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.8	5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	10.4	5.2	2.7	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	10.9	5	2.4	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9	10.3	5.2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	10.9	5	2.3	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	1.1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	1.1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.9	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	1.1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1.1	0.7	0.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.8	22.9	5.6	0.8	0	4.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.3	21.1	3.3	0.6	0	4.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	64.6	23.4	5.9	0.8	0	5.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.7	21.3	3.4	0.7	0	4.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.8	23.8	5.3	0.5	0	4.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.2	21	3.2	0.6	0	5
U2 - Efficacy additive alpha prior Beta(1, 1)	65	23.9	5.4	0.8	0	4.9
U3 - Efficacy additive alpha prior Beta(1, 1)	70.2	21.4	3.1	0.5	0	4.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.7	7.7	3.8	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	7.6	3.5	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	7.8	3.8	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.2	7.6	3.5	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.8	7.7	3.8	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	7.5	3.5	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.8	7.7	3.7	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.5	7.5	3.5	1.6	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.3	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.2	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.4	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.3	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.4	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.3	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.4	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.3	1	0.5	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	10.7	8.3	41.5	25.2	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.5	11.7	7.4	39.4	21.7	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.8	11.2	7.9	41.7	25	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	12	7.6	39.6	22	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	11.1	8.5	42.3	24.9	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18	12.8	7.6	39.1	22.3	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	14	9.8	8.3	42.4	25.2	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	18.5	12.9	7.3	38.2	22.9	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.8	5.3	9.3	6.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	4.9	5.1	9.5	5.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	4.8	5.3	9.3	6.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	4.8	5	9.5	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.8	5.2	9.2	6.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	4.9	5.1	9.5	5.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.8	5.3	9.3	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	4.8	5	9.5	5.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.5	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.5	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.5	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.4	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.5	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.5	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.5	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.4	2.3	1.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	45.5	17.1	7.4	2.2	14.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	45.3	15.3	5.5	1.6	15.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.9	42.9	18.7	7.3	1.9	15.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	43.9	15.6	6.4	1.4	15.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14	43.4	17.7	8.1	2	14.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	45.1	14.8	6.4	1.4	14.9
U2 - Efficacy additive alpha prior Beta(1, 1)	14	43.5	17.8	7.7	2.2	14.8
U3 - Efficacy additive alpha prior Beta(1, 1)	17.1	45.4	15.1	6.2	1.5	14.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	9.6	5.4	3.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	9.9	5.2	3.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	9.5	5.5	3.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	9.8	5.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	9.5	5.5	3.5	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	9.8	5.2	3.3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	9.7	5.4	3.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9	9.9	5.2	3.2	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.1	1.3	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4	1.4	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4	1.3	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4	1.3	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.3	1	0.3	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	15.6	13	32.4	17.4	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.4	16.8	11.8	28.4	11.7	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.8	14.6	13.4	32.9	16.6	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.2	17.5	11.2	28.6	11.7	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	15.8	13.4	31.6	17	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29	17.3	11.8	30	11.1	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	21.6	15.6	13.2	32.6	16.2	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	30.2	17.6	11	27.6	12.8	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	5.7	5.5	7.8	5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	5.7	5.3	7.9	4.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.7	5.6	7.8	5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	5.8	5.3	7.8	4.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	5.7	5.5	7.8	5.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.4	5.7	5.3	7.9	4.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	5.7	5.6	7.8	5	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.4	5.8	5.3	7.8	4.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.6	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.6	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.6	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.6	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.6	2.3	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.6	2.2	1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	51.9	26.3	13.6	5.6	1.1	1.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	60	25.6	9.9	2.4	0.3	1.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	51.4	27.9	12.3	5.6	1.1	1.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	59	26.1	10.4	2.5	0.4	1.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	50.5	27.7	13.2	6.1	1	1.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.9	25.1	10.7	2.3	0.4	1.6
U2 - Efficacy additive alpha prior Beta(1, 1)	50.4	27.7	13.8	5.4	1	1.7
U3 - Efficacy additive alpha prior Beta(1, 1)	60.3	25.4	9.7	2.4	0.5	1.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	8	5.9	4.4	1.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	7.8	5.5	4.2	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	8	5.9	4.4	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	7.8	5.5	4.3	1.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	8	5.9	4.5	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	7.8	5.5	4.2	1.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	8	5.9	4.5	1.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	7.8	5.5	4.3	1.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4	1.8	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.9	1.7	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4	1.9	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.9	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4	1.8	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.9	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4	1.9	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.9	1.7	1.6	0.5	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	43.1	19.9	17.2	12.3	7.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	48.4	22	14.6	9.7	5.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	43.7	21.1	15.6	12.1	7.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	48.3	22.6	14.7	9.9	4.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	44	20.9	15.2	11.8	8.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	49.7	21.5	13.5	10.2	5.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	43.7	20.6	16	11.6	8.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	48.2	22.2	15.2	9.5	4.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	4.7	6	8.2	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5	6.2	8.1	5.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	4.7	6.1	8.1	6.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	5	6.2	8.1	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.7	6.1	8.2	6.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5	6.2	8.2	5.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.7	6.1	8.2	6.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	5	6.3	8.2	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	2	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.9	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	2	2.9	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	2	2.9	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	2	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	2.1	2.9	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	2	2.9	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	2.1	2.9	1.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.5	56.1	18.1	3.1	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27	57.6	13.5	1.8	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.9	55.6	18.4	2.8	0.2	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	57.5	13.4	2.1	0.1	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.8	54.6	18.2	3	0.3	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.5	59	14.8	1.6	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	22.6	55.5	18.9	2.7	0.2	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	26.4	58.7	12.8	1.9	0	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	13.2	5.8	1.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	13.6	5.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	13.2	5.9	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	13.7	5.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	13.2	5.8	1.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	13.8	5.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	13.3	6	1.6	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	13.7	5.3	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1.1	0.4	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.7	7	9.1	15.4	55.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	11	9.5	11.3	48.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	8.2	8.9	14.3	56.1	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.5	11.1	9.8	11.4	48.2	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	8.2	9	15.5	55.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	10.5	9.5	11.6	49.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	12.7	7.8	9.8	15	54.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	20	11.5	8.8	12.1	47.6	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.9	5.1	6.5	11	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.7	4.1	5.1	6.6	10.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.9	5.1	6.5	11	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.1	5.1	6.6	10.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.9	5.1	6.6	10.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.1	5.1	6.6	10.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.9	5.1	6.5	10.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	4.1	5.1	6.6	10.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.5	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.5	2.5	1.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	58	19.7	2.7	0.1	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	58.5	16.3	2.8	0	2.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	57.6	20.2	2.7	0	2.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	58.4	16.7	2.5	0	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.7	57.4	20.9	2.8	0	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	58.1	17	2.2	0	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	17.5	55	22.6	2.7	0	2.2
U3 - Efficacy additive alpha prior Beta(1, 1)	20.4	57.2	17.8	2.3	0	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	12	7.1	2.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.9	12.3	6.6	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	12	7	2.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.9	12.3	6.6	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	12	7	2.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	12.2	6.7	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.5	12	7	2.8	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.9	12.2	6.6	2.6	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.5	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.4	1.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.4	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.4	1.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.5	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.4	1.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.5	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.4	1.6	0.9	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	43.1	28.8	13.4	2	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	47.1	23	10.2	1.5	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	42.4	30.1	13.1	2.1	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	46.8	23.8	10.1	1.9	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	43	28.6	13.9	2	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	47.8	24.1	9.7	1.5	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	12.6	43.7	28.1	13.5	1.7	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	17.1	48.7	21.9	10.3	1.6	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	9.2	8.2	5.7	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	9.6	7.7	5.4	2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	9.1	8.2	5.7	2.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	9.8	7.7	5.4	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	9.3	8.1	5.7	2.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	9.7	7.7	5.4	2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	9.1	8.1	5.8	2.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	9.7	7.8	5.4	2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	2.3	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	2.2	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	2.2	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.7	2.2	1.7	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	2.2	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	2.2	1.7	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.7	2.2	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	2.2	1.7	0.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	57.9	1.8	0	0	0	40.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.4	1.4	0	0	0	40.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	57.8	2	0	0	0	40.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	58.2	1.5	0	0	0	40.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.2	1.9	0	0	0	39.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58	1.4	0	0	0	40.6
U2 - Efficacy additive alpha prior Beta(1, 1)	58	1.8	0	0	0	40.2
U3 - Efficacy additive alpha prior Beta(1, 1)	58.1	1.8	0	0	0	40.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	2.7	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	2.6	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	2.8	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.2	2.7	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	2.7	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	2.7	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20	2.8	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.2	2.6	0.3	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	59	14.7	4.2	0.1	1.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.4	58.4	13.1	3.7	0	1.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	58.9	14.5	3.9	0.2	1.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.1	58.5	13.1	3.8	0	1.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	59.3	15.1	3.9	0.2	1.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.8	57.4	12.2	4	0	1.6
U2 - Efficacy additive alpha prior Beta(1, 1)	20.8	58.8	14.5	4.1	0.3	1.5
U3 - Efficacy additive alpha prior Beta(1, 1)	24.8	57.1	13.4	3.4	0	1.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	12.4	6.2	2.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	12.5	5.8	2.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	12.5	6.1	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	12.5	5.8	2.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	12.5	6.2	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	12.4	5.9	2.6	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.2	12.5	6.1	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	12.5	5.8	2.6	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.5	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.4	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.5	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.4	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.4	1.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	4.5	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.4	1.7	0.7	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.6	5.7	0.2	0	0	71.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	6.5	0.3	0	0	71
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23	5.7	0.2	0	0	71.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22	6.5	0.1	0	0	71.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.9	5	0.2	0	0	71.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.9	6.1	0.1	0	0	71.9
U2 - Efficacy additive alpha prior Beta(1, 1)	22.9	5.2	0.3	0	0	71.6
U3 - Efficacy additive alpha prior Beta(1, 1)	22.1	6.1	0.1	0	0	71.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.4	5.3	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	5.1	1.1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	5.3	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	5.1	1.1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.4	5.3	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	5.1	1.2	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.3	5.3	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	5.1	1.1	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.7	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.6	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.7	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.6	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.8	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.6	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.7	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.6	0.3	0.1	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.4	1.2	4.4	50.5	41.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	1.9	5.3	54.1	34.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.8	1.2	4	50	42	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	2	5.4	54.5	34	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.8	0.9	3.7	52.6	40	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	2	4.4	54.4	34.7	0
U2 - Efficacy additive alpha prior Beta(1, 1)	2.7	1.2	3.6	51.7	40.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	4.4	1.9	5.3	53.2	35.2	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.1	3.9	11.7	8.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	4	12.2	7.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	3.9	11.7	8.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	4	12.4	7.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.1	3.9	11.8	8.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	4	12.3	7.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.1	3.9	11.8	8.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	4	12.1	7.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.3	27.3	34.3	6.7	0.3	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	37.9	26.6	30.1	5.1	0.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.5	26.7	35.3	6.3	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.2	27.1	30.6	4.9	0.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.2	27.1	35.1	6.2	0.3	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.1	26.4	30	5.2	0.2	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	31.7	27	34.7	6.2	0.3	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	37.4	26.4	30.8	5.1	0.2	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	8.7	8.9	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	8.7	8.6	3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	8.6	8.9	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	8.8	8.6	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	8.7	8.9	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	8.7	8.4	3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	8.7	8.9	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	8.7	8.6	3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.6	1.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.6	1.8	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.6	1.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.6	1.8	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.6	1.8	1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.6	1.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.6	1.8	1	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.7	29.9	33.4	6.9	0.1	3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.9	30.6	29.3	6.2	0	3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26	30.4	34.2	6.4	0.1	2.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.1	30.2	29.9	5.9	0	2.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.1	29.1	33.9	7.1	0	2.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.2	31.7	27.8	6.4	0.1	2.8
U2 - Efficacy additive alpha prior Beta(1, 1)	25.4	30.4	34	7.1	0.1	3
U3 - Efficacy additive alpha prior Beta(1, 1)	32.4	30.5	28.9	5.5	0	2.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	8.6	8.6	3.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	8.7	8.3	3.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.3	8.6	8.6	3.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	8.6	8.4	3.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	8.5	8.7	3.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	8.7	8.2	3.4	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.3	8.5	8.6	3.7	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	8.7	8.3	3.4	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.7	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.8	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	1.7	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	1.7	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.8	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	1.7	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.8	1.6	1.2	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.8	26.6	45.9	9.2	0.8	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.1	26.3	40.3	9	0.6	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	24	46.9	10.1	0.8	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.1	26.4	40.4	8.8	0.5	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.1	23.8	48	9.9	0.6	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.6	26.4	39.5	9.1	0.6	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.2	24.6	46.1	10.7	0.7	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	23.8	25	40.4	9.7	0.4	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.3	7.7	10.2	4.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	7.8	10.1	4.3	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.3	7.7	10.1	4.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.9	7.9	9.9	4.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.3	7.7	10.3	4.5	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.8	8	10	4.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.3	7.7	10.2	4.5	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.8	7.9	10	4.2	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	2	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.6	2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	2.1	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.6	2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	2	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	2.1	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	2	1.4	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	51.4	9.1	0.9	0	17.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22	51.8	8.3	0.5	0	17.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	53.3	8.6	0.9	0	17.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	50.8	8.5	0.5	0	17.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.1	51.6	9.6	0.6	0	17.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.9	51.9	8.1	0.4	0	17.7
U2 - Efficacy additive alpha prior Beta(1, 1)	20.1	53.3	9.4	0.7	0	16.5
U3 - Efficacy additive alpha prior Beta(1, 1)	21.4	51.4	8.5	0.5	0	18.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	12	4.5	1.5	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	11.8	4.3	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	11.9	4.5	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	11.8	4.3	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	11.8	4.5	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	11.9	4.2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	12	4.5	1.5	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	11.8	4.2	1.4	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.3	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.4	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.4	1.2	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.5	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4.4	1.2	0.5	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.7	26.9	4.9	0.2	0	2.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	70	24.5	3.2	0.1	0	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.2	26.3	4	0.3	0	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69	26	2.7	0.1	0	2.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68	25.6	4	0.1	0	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	68	26.8	2.7	0.2	0	2.3
U2 - Efficacy additive alpha prior Beta(1, 1)	66	27.2	4.5	0.1	0	2.2
U3 - Efficacy additive alpha prior Beta(1, 1)	69.4	25.6	2.3	0.2	0	2.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	8.6	2.9	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	8.2	2.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	8.5	2.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.1	8.4	2.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	8.4	2.9	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18	8.4	2.4	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.3	8.6	2.9	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.1	8.4	2.4	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.9	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.9	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.9	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.9	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	11.1	13.3	40.5	16.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.1	12.4	12.1	37.3	10.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19	10.9	13.5	40.8	15.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28	12.1	10.9	37.5	11.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	10.9	13.8	40.9	15.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.6	13.3	11.2	36.9	11	0
U2 - Efficacy additive alpha prior Beta(1, 1)	19.9	10.6	13.2	40.7	15.6	0
U3 - Efficacy additive alpha prior Beta(1, 1)	28	12.8	11.8	36.3	11.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	4.6	5.9	10	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.7	5.9	10.3	4.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	4.6	5.9	10	5.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.7	5.9	10.2	4.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	4.6	5.9	10.1	5.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.7	5.8	10.2	4.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.3	4.5	5.9	10.1	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.7	5.7	10.2	4.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.7	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.7	2.3	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.6	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.7	2.3	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.7	2.4	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.6	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.6	2.4	1.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.5	31.7	27.7	28	7.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.1	37.4	27	22.6	4.9	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	32.6	27.2	27.8	7.1	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.9	37.5	26.9	22.8	4.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	31.6	27.7	27.2	7.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	37.8	26.1	23.6	4.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	32.4	26.8	28.1	6.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	7.8	37.5	27.4	22.2	5.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	9.7	7.5	6.4	2.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	10.7	7.5	5.9	2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	9.8	7.5	6.4	2.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	10.7	7.6	5.8	2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	9.8	7.6	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	10.9	7.4	5.9	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	9.9	7.4	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	10.9	7.4	5.8	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1.5	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.6	1.5	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.6	1.4	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.4	1.2	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.6	1.4	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	1.2	0.5	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	12.9	33.7	24.7	13.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	15.2	33.9	21.7	8.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.7	12.6	33.5	25.7	12.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	15.7	33.1	22.2	8.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	12.4	34.5	24.8	12.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.7	14.6	34.4	22.1	8.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	15.7	12.5	34	25.1	12.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	20.6	13.9	36.1	21.4	8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4	8.3	8.4	5.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.4	8.6	8.3	5.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4	8.3	8.4	5.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.3	8.6	8.3	5.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4	8.3	8.4	5.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.3	8.7	8.3	5.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4	8.4	8.4	5.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.6	4.3	8.7	8.3	5.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.9	2.7	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.9	2.7	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.8	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.8	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.9	2.7	1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	4.1	13.3	40	32.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.2	6	12.6	39.4	26.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	4.6	12.9	39.2	32.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	6.3	11.8	40.3	26.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	4.6	12.6	39.9	32.6	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.2	6.2	12.2	39.8	26.6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	10.2	4.7	12.2	40.1	32.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15.7	6.2	12.4	39.1	26.6	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	3.5	5.8	9.7	7.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.8	10.2	6.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.5	5.7	9.7	7.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.7	5.7	10.2	6.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.6	5.7	9.8	7.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.7	5.8	10	7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.5	5.7	9.8	7.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.8	10.1	6.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1.4	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.6	1.4	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.6	1.5	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.7	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	12.7	44.1	20	6.5	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.9	14	41.7	18.2	4	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	12.2	44.3	21	6.5	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.7	14.4	41.5	18.9	4.3	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.6	11.8	45.3	21.1	5.9	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	13.8	41.7	18.2	3.8	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	15.2	12.2	45.3	20.4	6.6	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	22.6	14	40.4	19	3.8	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	6	9.1	6.6	3.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	6	9.1	6.3	3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	6	9.2	6.5	3.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	6.1	9.1	6.4	2.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	5.9	9.3	6.5	3.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	6	9.1	6.3	3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	5.9	9.2	6.6	3.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.6	6	9.2	6.3	2.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	2.1	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.3	2	1.9	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.3	2.1	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.4	2	2	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.3	2	2.1	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	2	1.9	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.3	2	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.3	2	1.9	0.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	37.3	31.3	15.6	5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.7	43.3	26.2	12	2.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	36.6	30.5	16.1	5.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.9	42.4	26.4	11.1	3.2	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	36.6	31.4	16.1	5.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.6	43.2	26.7	11.4	3.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	38.3	30.3	16.3	4.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15.4	43.3	27.6	11	2.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	9.9	8.5	5.1	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	11.1	8.1	4.4	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	10.1	8.4	4.9	2.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	11.1	8.2	4.4	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	10.1	8.4	4.9	2.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	11.1	8.2	4.4	1.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	10.3	8.4	4.9	2.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	11.2	8.3	4.3	1.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.8	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.8	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.8	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.8	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.8	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.8	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.8	1.2	0.3	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	44.1	16.7	9.3	3.4	4.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.5	43.9	14.6	8	1.8	5.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.6	43.3	17.4	9.4	3.6	4.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.3	43	13.8	8.2	1.8	4.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.9	45.1	16.4	9	3.3	4.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.7	44.3	14.6	7.6	1.8	5
U2 - Efficacy additive alpha prior Beta(1, 1)	20.9	46.3	16.5	9	3.2	4.1
U3 - Efficacy additive alpha prior Beta(1, 1)	27.2	44.1	13.2	8.4	1.5	5.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	9.7	5.9	4.1	1.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.6	9.6	5.7	3.9	1.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.9	9.7	6	4.1	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	9.6	5.8	4	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	9.8	5.9	4.2	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	9.6	5.8	3.9	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	8	9.7	5.9	4	1.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	9.8	5.7	3.9	1.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.3	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.3	1.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.3	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.3	1.6	1.3	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.3	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.3	1.6	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.4	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.3	1.6	1.2	0.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.9	51.9	16.8	4.8	1.1	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.1	52.2	12.1	3.5	0.5	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.5	52.6	15.3	5.3	0.9	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.1	52.1	12.9	3.9	0.4	2.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.4	50.7	17.4	5.4	0.8	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.3	53.1	12.9	3.7	0.4	2.6
U2 - Efficacy additive alpha prior Beta(1, 1)	23.9	51.3	16.1	5.4	0.9	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	28.3	51.4	13.5	3.9	0.5	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	10.9	6.2	3.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	10.8	5.9	3.5	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	10.9	6.1	3.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	10.8	5.9	3.5	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	10.8	6.2	3.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	10.8	5.8	3.5	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.8	10.8	6.2	3.7	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	10.8	5.8	3.5	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.7	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.6	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.7	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.6	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.7	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.6	1.6	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.7	1.7	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4.5	1.6	1.2	0.3	